

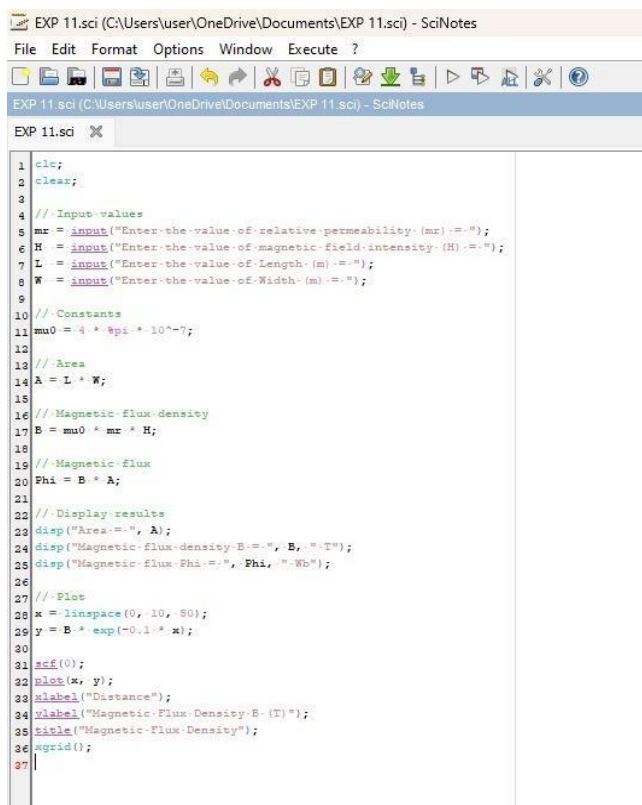
EXPERIMENT NO 11

Determination and plotting of Magnetic Flux density in Ferromagnetic Materials with different (i) length, (ii) width and (iii) magnetic field intensity.

Aim:

To write program for Magnetic flux density in ferromagnetic materials using SCILAB. **Software required:**

•SCILAB version 6.0.1 **Program:**



```
EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
EXP 11.sci X
1 clear;
2 clear;
3
4 // Input values
5 mr = input("Enter the value of relative permeability (mr) :-");
6 H = input("Enter the value of magnetic field intensity (H) :-");
7 L = input("Enter the value of Length (m) :-");
8 W = input("Enter the value of Width (m) :-");
9
10 // Constants
11 mu0 = 4 * pi * 10^-7;
12
13 // Area
14 A = L * W;
15
16 // Magnetic flux density
17 B = mu0 * mr * H;
18
19 // Magnetic flux
20 Phi = B * A;
21
22 // Display results
23 disp("Area :- ", A);
24 disp("Magnetic flux density B :- ", B, " T");
25 disp("Magnetic flux Phi :- ", Phi, " Wb");
26
27 // Plot
28 x = linspace(0, 10, 50);
29 y = B * exp(-0.1 * x);
30
31 clf();
32 plot(x, y);
33 xlabel("Distance");
34 ylabel("Magnetic Flux Density B (T)");
35 title("Magnetic Flux Density");
36 xgrid();
37
```

Output:

Scilab 2026.1.0 Console

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File Browser

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- ex 6.sc
- EXP 11.sc

File/directory filter

Scilab 2026.1.0 Console

Enter the value of relative permeability (μ_r) = 3×10^{-7}

Enter the value of magnetic field intensity (H) = 6

Enter the value of Length (m) = 4

Enter the value of Width (m) = 2

```

"Area = "
8.
"Magnetic flux density B = "
0.0001734
" T"
"Magnetic flux Phi = "
0.0013873
" Wb"
-->

```

Variable Browser

Name	Value	Type	Visibility	Memory
A	8	Double	local	216 B
B	0.000173	Double	local	216 B
H	6	Double	local	216 B
L	4	Double	local	216 B
Phi	0.00139	Double	local	216 B
W	2	Double	local	216 B
ans	1x1 Graphic ...		local	216 B
mu_r	23	Double	local	216 B
mu0	1.26e-06	Double	local	216 B

Command History

```

-- 10/08/2026 13:50:17 -- //
10
5*10^-6
10
-- 10/08/2026 14:33:01 -- //
-- 10/08/2026 14:33:17 -- //
-- 10/08/2026 14:50:34 -- //
-- 24/08/2026 04:27:38 -- //
-- 10/08/2026 04:29:27 -- //
3*10^-7
6
4
2

```

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Notification Suggestions

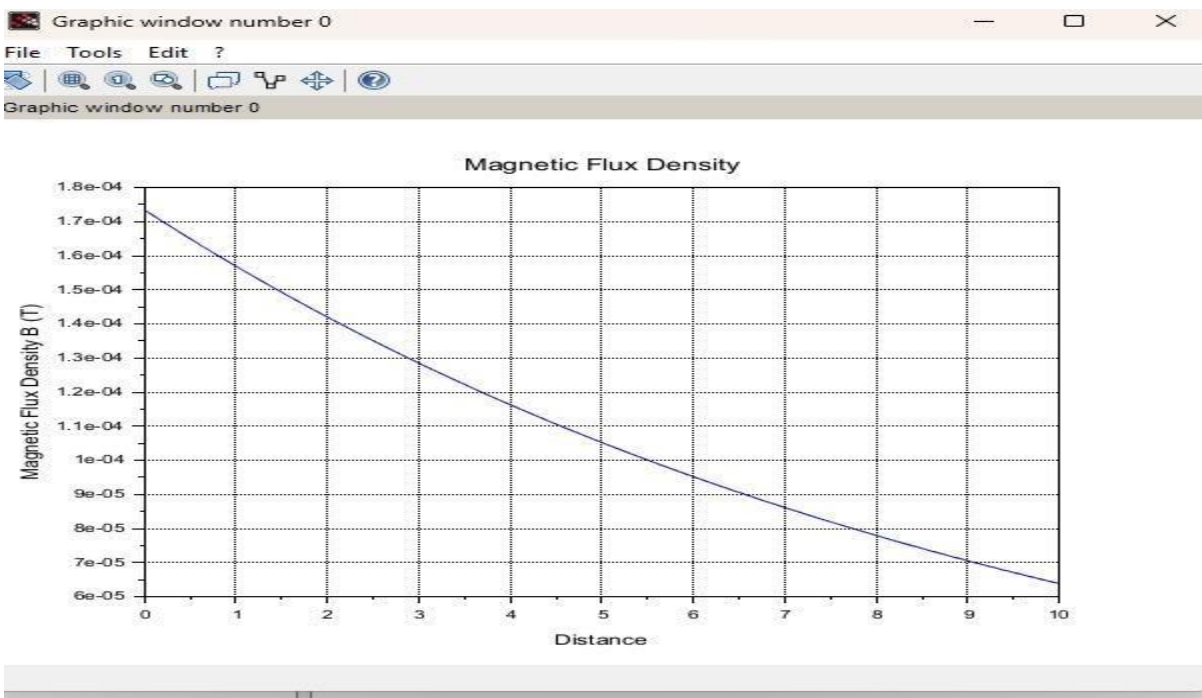
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27°

ENG IN

4:34 AM 8/24/2026



Theoretical calculations:

Exp: 11

$$nr = 5$$

$$H = 3 \text{ cm}$$

$$L = 8 \text{ cm}$$

$$W = 2 \text{ mm}$$

• constant & magnetic flux (B):

~~μ_0~~ ~~μ_r~~

$$\mu_0 = 4 \times 3.14 \times 10^{-7} = 1.256 \times 10^{-6}$$

$$= 4 \times 3.14 \times 10^{-7} = 1.256$$

$$= \mu_0 \cdot nr \cdot H = (1.256 \times 10^{-6}) \times 5 \times 3 = 1.884 \times 10^{-5} \text{ T}$$

Area (A)

$$A = 0.06 \times 0.002 = 1.2 \times 10^{-4} \text{ m}^2$$

$$\text{Flux density} = \frac{1.884 \times 10^{-5}}{1.2 \times 10^{-4}} = 0.157$$

Case 1: Examine the effect of length of a ferromagnetic material on the magnetic flux density (B).

Analytical :

Calculate the magnetic flux density for the following conditions

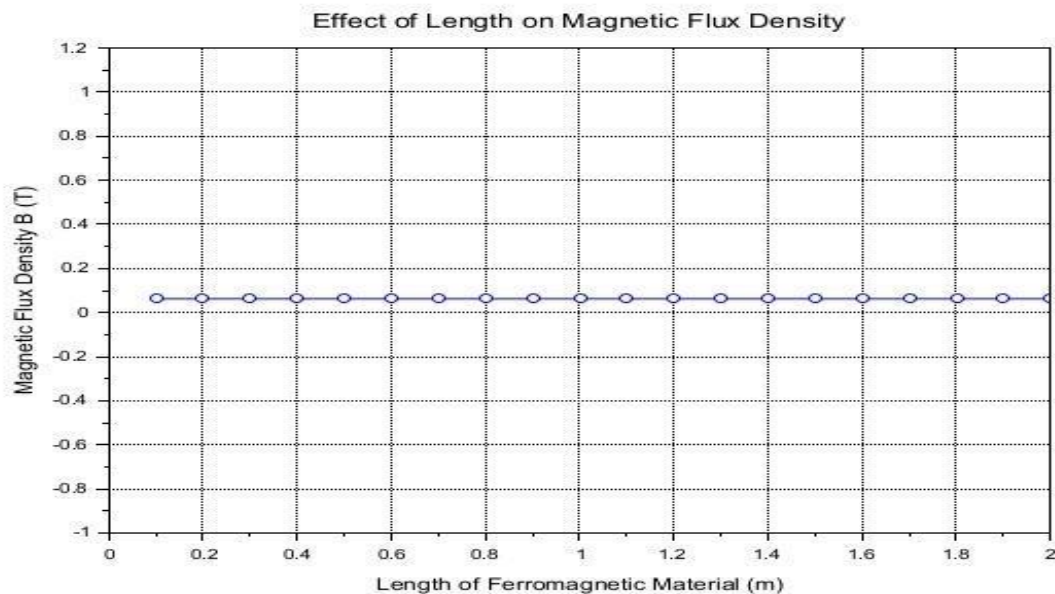
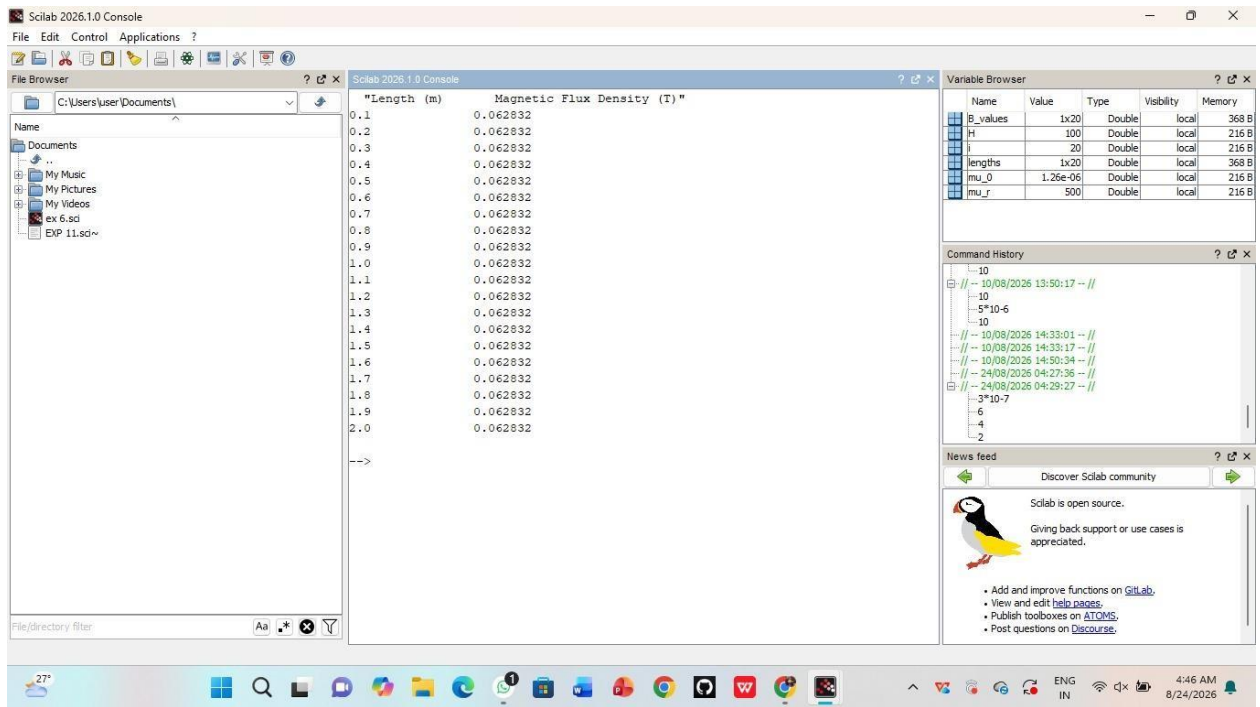
- The material is ferromagnetic with relative permeability (μ_r) of 500.
- The magnetic field intensity (H) is held constant at 100 A/m.
- The length of the material varies from 0.1 m to 2 m.

Program:

```
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EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
EXP 11.sci X

1 |clc;
2 |clear;
3
4 //Constants
5 mu_r = 500; //Relative permeability
6 H = 100; //Magnetic field intensity (A/m)
7 lengths = 0.1:0.1:2; //Lengths from 0.1 m to 2 m
8 mu_0 = 4*pi*10^(-7);
9
10 //Initialize array
11 B_values = zeros(1, length(lengths));
12
13 //Calculate magnetic flux density
14 for i = 1:length(lengths)
15     B_values(i) = mu_r * mu_0 * H;
16 end
17
18 //Display results
19 disp('Length (m) -----Magnetic Flux Density (T)');
20
21 for i = 1:length(lengths)
22     fprintf('%0.1f ----- %0.6f\n', lengths(i), B_values(i));
23 end
24
25 //Plot
26 clf;
27 plot(lengths, B_values, '-o');
28
29 xlabel('Length of Ferromagnetic Material (m)');
30 ylabel('Magnetic Flux Density B (T)');
31 title('Effect of Length on Magnetic Flux Density');
32
33 xgrid();
34
```

Output:



Case 2: To investigate how the magnetic flux density (B) in a ferromagnetic material changes with

Different widths at a constant length and magnetic field intensity.

Analytical:

Calculate magnetic flux density (B) in a ferromagnetic material changes with different widths at a constant length And magnetic field intensity for the following conditions

- Material: Soft iron (same as in Case Study 1)
- Length (L): Constant at 10 cm
- Magnetic Field Intensity (H): Constant at 1000 A/m
- Width (W): Varies from 2 cm to 10 cm (i.e., 2 cm, 4 cm, 6 cm, 8 cm, 10 cm)

Program:

```
EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
EXP 11.sci x
1 clear;
2 clear;
3
4 //Constants
5 L = 0.1; //Length in meters (10 cm)
6 H = 1000; //Magnetic field intensity in A/m
7 mu_0 = 4 * pi * 10^(-7); //Permeability of free space (H/m)
8
9 //Widths in meters (2 cm to 10 cm)
10 widths = [0.02, 0.04, 0.06, 0.08, 0.10];
11 num_widths = length(widths);
12
13 //Relative permeability of soft iron
14 mu_r = 1000;
15
16 //Initialize array
17 B_values = zeros(1, num_widths);
18
19 //Calculate magnetic flux density
20 for i = 1:num_widths
21     B_values(i) = mu_0 * mu_r * H;
22 end
23
24 //Display results
25 disp("Width (m) ---- Flux Density (T)");
26
27 for i = 1:num_widths
28     fprintf("Width: %f ---- Flux Density: %f\n", widths(i), B_values(i));
29 end
30
31 //Plot the results
32 clf;
33 plot(widths, B_values, "o");
34 xlabel("Width of Ferromagnetic Material (m)");
35 ylabel("Magnetic Flux Density B (T)");
36 title("Effect of Width on Magnetic Flux Density");
37 grid();
38
```

Output:

Scilab 2026.1.0 Console

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EXP 11.sc

Scilab 2026.1.0 Console

Width (m)	Flux Density (T)
0.02	1.256637
0.04	1.256637
0.06	1.256637
0.08	1.256637
0.10	1.256637

-->

Variable Browser

Name	Value	Type	Visibility	Memory
B_values	1x5	Double	local	248 B
H	1e+03	Double	local	216 B
L	0.1	Double	local	216 B
i	5	Double	local	216 B
mu_0	1.26e-06	Double	local	216 B
mu_r	1e+03	Double	local	216 B
num_widths	5	Double	local	216 B
widths	1x5	Double	local	248 B

Command History

```
// -- 10/08/2026 13:50:17 -- //
10
10
5*10^-6
10
// -- 10/08/2026 14:33:01 -- //
// -- 10/08/2026 14:33:17 -- //
// -- 10/08/2026 14:50:34 -- //
// -- 24/08/2026 04:27:38 -- //
3*10^-7
6
4
2
```

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Giving back support or use cases is appreciated.

- Add and improve functions on [GitLab](#).
- View and edit [help pages](#).
- Publish toolboxes on [ATOS](#).
- Post questions on [Discourse](#).

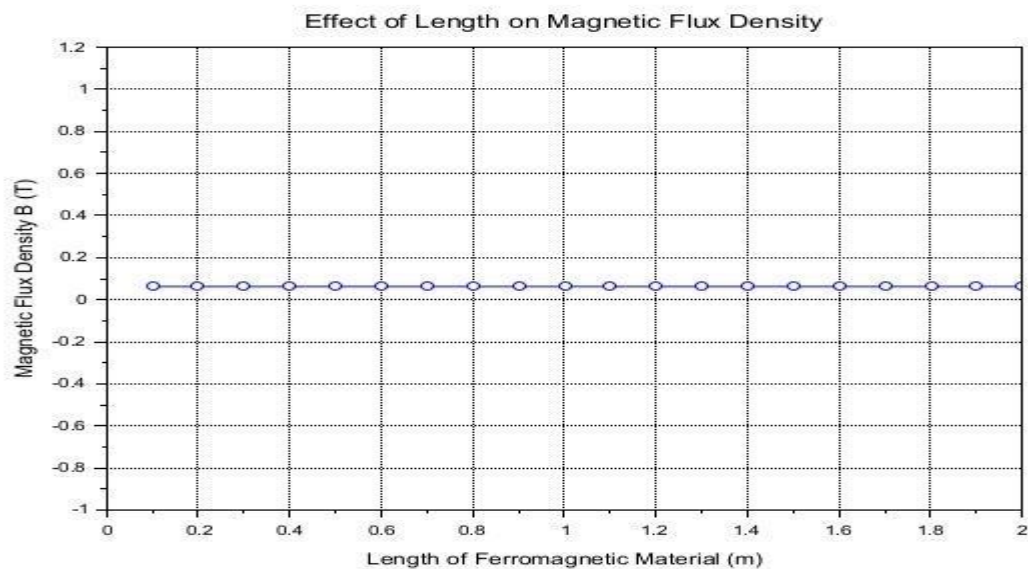
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Graphic window number 0

File Tools Edit ?

Graphic window number 0



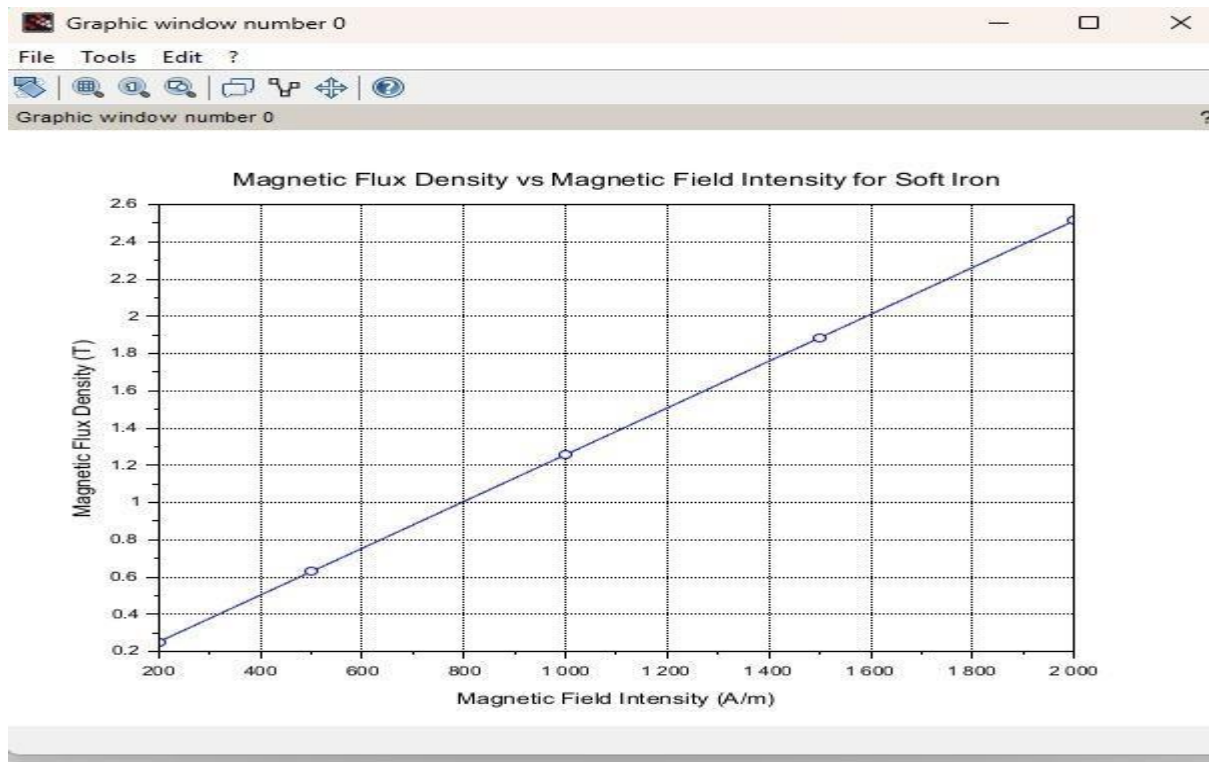
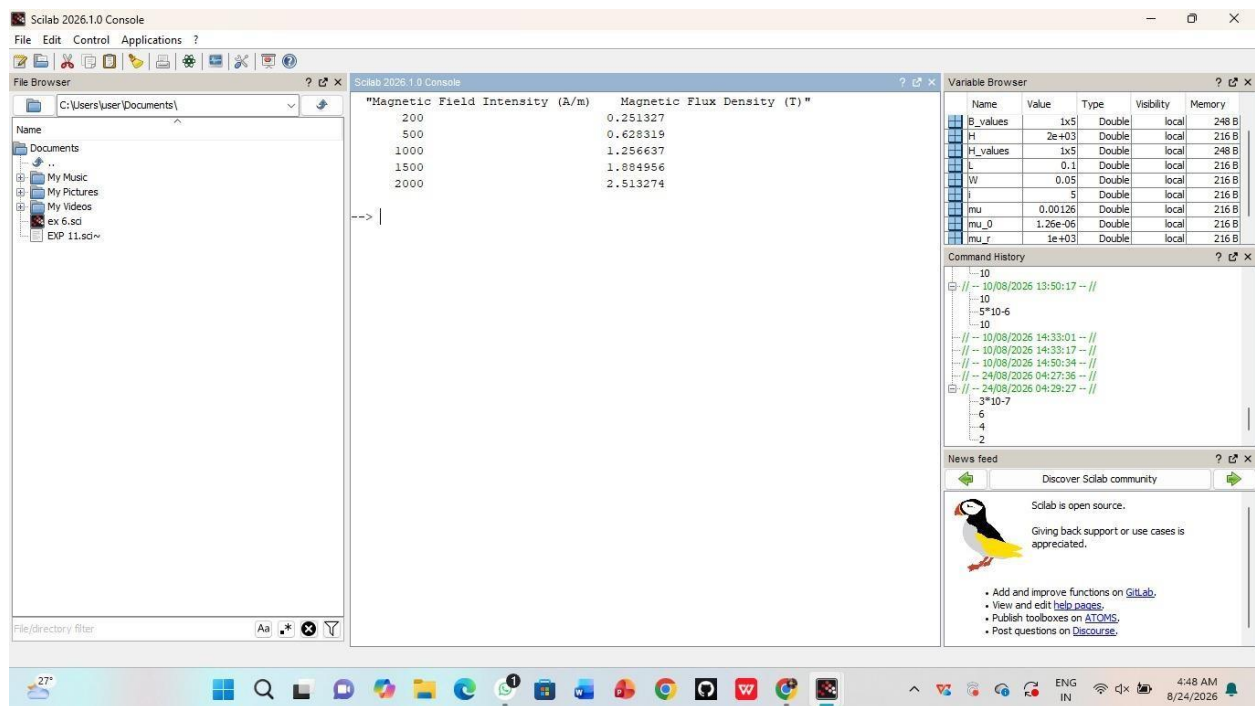
Case 3: To examine how the magnetic flux density (B) in a ferromagnetic material change with different values of magnetic field intensity at constant length and width.

Program:

```
EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
EXP 11.sci X

3
4 //Define constants
5 L = 0.1; //Length in meters (10 cm)
6 W = 0.05; //Width in meters (5 cm)
7
8 H_values = [200, 500, 1000, 1500, 2000]; //H in A/m
9
10 mu_0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
11 mu_r_initial = 1000; //Relative permeability of soft iron
12
13 //Initialise array
14 B_values = zeros(1, length(H_values));
15
16 //Calculate B for each H value
17 for i = 1:length(H_values)
18     H = H_values(i);
19     mu_r = mu_r_initial;
20
21     //Calculate absolute permeability
22     mu = mu_0 * mu_r;
23
24     //Calculate magnetic flux density
25     B_values(i) = mu * H;
26 end
27
28 //Display results
29 disp("Magnetic Field Intensity (A/m) --- Magnetic Flux Density (T)");
30
31 for i = 1:length(H_values)
32     fprintf("%10.0f-----%.6f\n", H_values(i), B_values(i));
33 end
34
35 //Plotting the results
36 clf();
37 plot(H_values, B_values, "-o");
38
39 xlabel("Magnetic Field Intensity (A/m)");
40 ylabel("Magnetic Flux Density (T)");
41 title("Magnetic Flux Density vs Magnetic Field Intensity for Soft Iron");
42
43 xgrid();
```


Output:



Result:

The effect of Magnetic field intensity of a ferromagnetic material on the magnetic flux density (B) is examined.